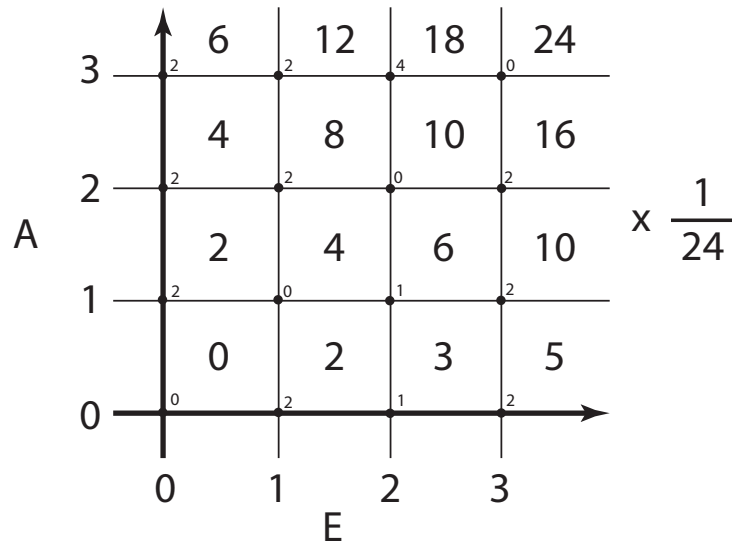


Rensselaer Polytechnic Institute
Department of Electrical, Computer, and Systems Engineering
ECSE 2500: Engineering Probability, Spring 2026
Homework #6 Solutions

1. (a) Each value of E in $[0, 3]$ has a $\frac{1}{4}$ probability of occurring since the distribution is uniform. We then have information about the conditional probability of A given E . For example, $P(A = 3|E = 2) = \frac{2}{3}$ and $P(A = 3, E = 2) = P(A = 3|E = 2)P(E = 2) = \frac{2}{3} \cdot \frac{1}{4} = \frac{1}{6}$. We can summarize the joint PMF values in the following table:

		E			
		0	1	2	3
A	0	0	$\frac{1}{12}$	$\frac{1}{24}$	$\frac{1}{12}$
	1	$\frac{1}{12}$	0	$\frac{1}{24}$	$\frac{1}{12}$
	2	$\frac{1}{12}$	$\frac{1}{12}$	0	$\frac{1}{12}$
	3	$\frac{1}{12}$	$\frac{1}{12}$	$\frac{1}{6}$	0

- (b) The joint CDF looks like a staircase function that's easiest to represent as a grid. Remember that the joint CDF is the sum of all probability to the left and below a given point (indicated by small numbers at the dots); we accumulate probability every time we cross a line.



- (c) From the joint PMF in part (a), it's easy to compute the marginal PMF of A just by summing up the rows; we obtain

x	$p_A(x)$
0	$\frac{5}{24}$
1	$\frac{5}{24}$
2	$\frac{1}{4}$
3	$\frac{1}{3}$

As expected these probabilities sum to 1, since the marginal is a valid PMF.

2. (a) Let's integrate this function and see what we get:

$$\begin{aligned}
 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} g(x, y) \, dx \, dy &= k \int_0^1 \int_0^1 x(1-x)y \, dx \, dy \\
 &= k \left(\frac{1}{2}x^2 - \frac{1}{3}x^3 \right) \Big|_{x=0}^{x=1} \left(\frac{1}{2}y^2 \right) \Big|_{y=0}^{y=1} \\
 &= k \left(\frac{1}{6} \right) \left(\frac{1}{2} \right) \\
 &= \frac{k}{12}
 \end{aligned}$$

Since for a valid PDF we need this integral to equal 1, this means that $k = 12$.

- (b) The joint CDF in the “interesting” range $x \in [0, 1], y \in [0, 1]$ is computed as

$$\begin{aligned}
 F_{XY}(x, y) \, dx \, dy &= \int_0^x \int_0^y f_{XY}(x, y) \, dx \, dy \\
 &= 12 \left(\frac{1}{2}x^2 - \frac{1}{3}x^3 \right) \left(\frac{1}{2}y^2 \right) \\
 &= (3x^2 - 2x^3)y^2
 \end{aligned}$$

- (c) To compute the marginals, we integrate out the variable we don't care about. For $x \in [0, 1]$ we have

$$\begin{aligned}
 f_X(x) &= \int_0^1 12x(1-x)y \, dy \\
 &= (12x(1-x)) \left(\frac{1}{2}y^2 \right) \Big|_{y=0}^{y=1} \\
 &= (12x(1-x)) \left(\frac{1}{2} \right) \\
 &= 6x(1-x) \quad x \in [0, 1], 0 \text{ otherwise}
 \end{aligned}$$

- (d) Similarly for $y \in [0, 1]$ we have

$$\begin{aligned}
 f_Y(y) &= \int_0^1 12x(1-x)y \, dx \\
 &= y \left(6x^2 - 4x^3 \right) \Big|_{x=0}^{x=1} \\
 &= y(2) \\
 &= 2y \quad y \in [0, 1], 0 \text{ otherwise}
 \end{aligned}$$

- (e) Yes, X and Y are independent since we can see that

$$\begin{aligned}
 f_{XY}(x, y) &= 12x(1-x)y \\
 &= (6x(1-x))(2y) \\
 &= f_X(x) f_Y(y)
 \end{aligned}$$

Also a clear sign of independence is that we've been able to decouple the x and y integrals throughout the problem.

(f)

$$\begin{aligned}P(Y < \sqrt{X}) &= \int_0^1 \int_0^{\sqrt{x}} 12x(1-x)y \, dy \, dx \\&= \int_0^1 (12x(1-x)) \left(\frac{1}{2}y^2 \right) \Big|_{y=0}^{y=\sqrt{x}} dx \\&= \int_0^1 (12x(1-x)) \left(\frac{1}{2}x \right) dx \\&= \int_0^1 6x^2(1-x) \, dx \\&= 2x^3 - \frac{6}{4}x^4 \Big|_{x=0}^{x=1} \\&= \frac{1}{2}\end{aligned}$$

3. (a) The correlation of X and Y is defined as $E(XY)$.

$$\begin{aligned}E(XY) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} xy f_{XY}(x, y) \, dx \, dy \\&= \frac{6}{19} \int_0^2 \int_0^1 xy(x^2 + y^3) \, dy \, dx = \frac{6}{19} \int_0^2 \left(\frac{1}{2}x^3y^2 + \frac{1}{5}xy^5 \right) \Big|_{y=0}^{y=1} dx \\&= \frac{6}{19} \int_0^2 \left(\frac{1}{2}x^3 + \frac{1}{5}x \right) dx \\&= \frac{6}{19} \left(\frac{1}{8}x^4 + \frac{1}{10}x^2 \right) \Big|_{x=0}^{x=2} \\&= \frac{6}{19} \cdot \frac{12}{5} = \frac{72}{95}\end{aligned}$$

- (b) The covariance of X and Y is defined as $E(XY) - E(X)E(Y)$, which means we first have to compute the marginal means $E(X)$ and $E(Y)$.

$$\begin{aligned}
 E(X) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x f_{XY}(x, y) \, dx \, dy \\
 &= \frac{6}{19} \int_0^2 \int_0^1 x(x^2 + y^3) \, dy \, dx = \frac{6}{19} \int_0^2 \left. x^3 y + \frac{1}{4} x y^4 \right|_{y=0}^{y=1} dx \\
 &= \frac{6}{19} \int_0^2 x^3 + \frac{1}{4} x \, dx \\
 &= \frac{6}{19} \left(\frac{1}{4} x^4 + \frac{1}{8} x^2 \right) \Big|_{x=0}^{x=2} \\
 &= \frac{6}{19} \cdot \frac{9}{2} = \frac{27}{19}
 \end{aligned}$$

$$\begin{aligned}
 E(Y) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y f_{XY}(x, y) \, dx \, dy \\
 &= \frac{6}{19} \int_0^2 \int_0^1 y(x^2 + y^3) \, dy \, dx = \frac{6}{19} \int_0^2 \left. \frac{1}{2} x^2 y^2 + \frac{1}{5} y^5 \right|_{y=0}^{y=1} dx \\
 &= \frac{6}{19} \int_0^2 \frac{1}{2} x^2 + \frac{1}{5} \, dx \\
 &= \frac{6}{19} \left(\frac{1}{6} x^3 + \frac{1}{5} x \right) \Big|_{x=0}^{x=2} \\
 &= \frac{6}{19} \cdot \frac{26}{15} = \frac{52}{95}
 \end{aligned}$$

Thus

$$\text{Cov}(X, Y) = \frac{72}{95} - \frac{27}{19} \cdot \frac{52}{95} = \frac{-36}{1805}$$

- (c) No, X and Y are not uncorrelated since $\text{Cov}(XY) \neq 0$.
 (d) Now we use the properties of expected value and the numbers we already computed:

$$\begin{aligned}
 E(95(X(1+Y) + 2Y(1-X))) &= 95E(X + 2Y - XY) \\
 &= 95(E(X) + 2E(Y) - E(XY)) \\
 &= 95\left(\frac{27}{19} + \frac{104}{95} - \frac{72}{95}\right) = 167
 \end{aligned}$$